Daisyworld, land and ocean: a guide to exploring the app

EC2113 Intro to Climate · Ocean module · app: joymonteiro.github.io/intro_climate_science/tutorials/daisyworld-land-ocean

This handout goes with the app "Daisyworld, land and ocean". Open the app next to it. The app runs in any browser, including on a phone, and needs no installation.

- **Part 1** explains what the model is and how to use the app. It has no equations.
- **Part 2** walks through ten runs. For each one it says what to set, shows the screen, asks you to predict what will happen, and then explains what you see. The equations come in here, one at a time, where they explain something on the screen. Try to answer each "Before you look" question before you read on.
- **Part 3** sums up what to take away, and says what the model leaves out.

You need no maths beyond the energy balance from earlier in the course.

> **The point in two sentences.** The ocean does two things to a planet: it makes it darker, and it makes it slower. Neither one gives the planet control over its temperature: more ocean means the daisies have less to work with, and the planet takes longer to respond to any change.

Part 1. The model and the app

1.1 Daisyworld in one paragraph

Daisyworld is a thought experiment by James Lovelock and Andrew Watson (1983). An imaginary planet circles a sun that slowly brightens, as our own sun has over billions of years. The only life is two kinds of daisy, black and white. Black daisies absorb sunlight and warm their surroundings; white daisies reflect it and cool theirs. Neither daisy "aims" to control anything. Each simply grows where it is comfortable. Yet together they hold the planet near the temperature they like over a wide range of sun, while a planet without them heats steadily. The model was built to show that life and climate can steer each other without anyone planning it. It is now a standard way to teach feedbacks.

1.2 What this version adds: an ocean

This app gives the planet an ocean, and the ocean does two jobs.

1. **It takes away the daisies' room, and it is dark.** Daisies can't grow on water, so they can only change the colour of the land. The ocean reflects little sunlight, less than anything on land, and its colour never changes. The more ocean, the less of the planet's colour the daisies control.
2. **It stores a lot of heat.** Warming water takes far more energy than warming land and air. The more ocean, the more slowly the planet warms or cools when the sun changes.

Those two jobs give the two questions this handout explores: **how much control** can the daisies have, and **how fast** does the planet respond?

1.3 What is in the model, in words

- **One temperature for the whole planet.** There are no places, no map and no seasons. Land and ocean share the same temperature.
- **The sun.** Its brightness is called L. $L = 1$ is the reference sun of the original model. The planet absorbs part of the sunlight and radiates heat to space. Its temperature settles where the two balance.
- **The surface.** It is made of four kinds of surface. Each has its own **albedo**, the fraction of sunlight it reflects:

Surface	Albedo	Effect
White daisies	0.75	reflect most sunlight: cool the planet
Bare ground	0.50	in between
Black daisies	0.25	absorb most sunlight: warm the planet
Ocean	0.10 (adjustable)	darker than anything on land, and fixed

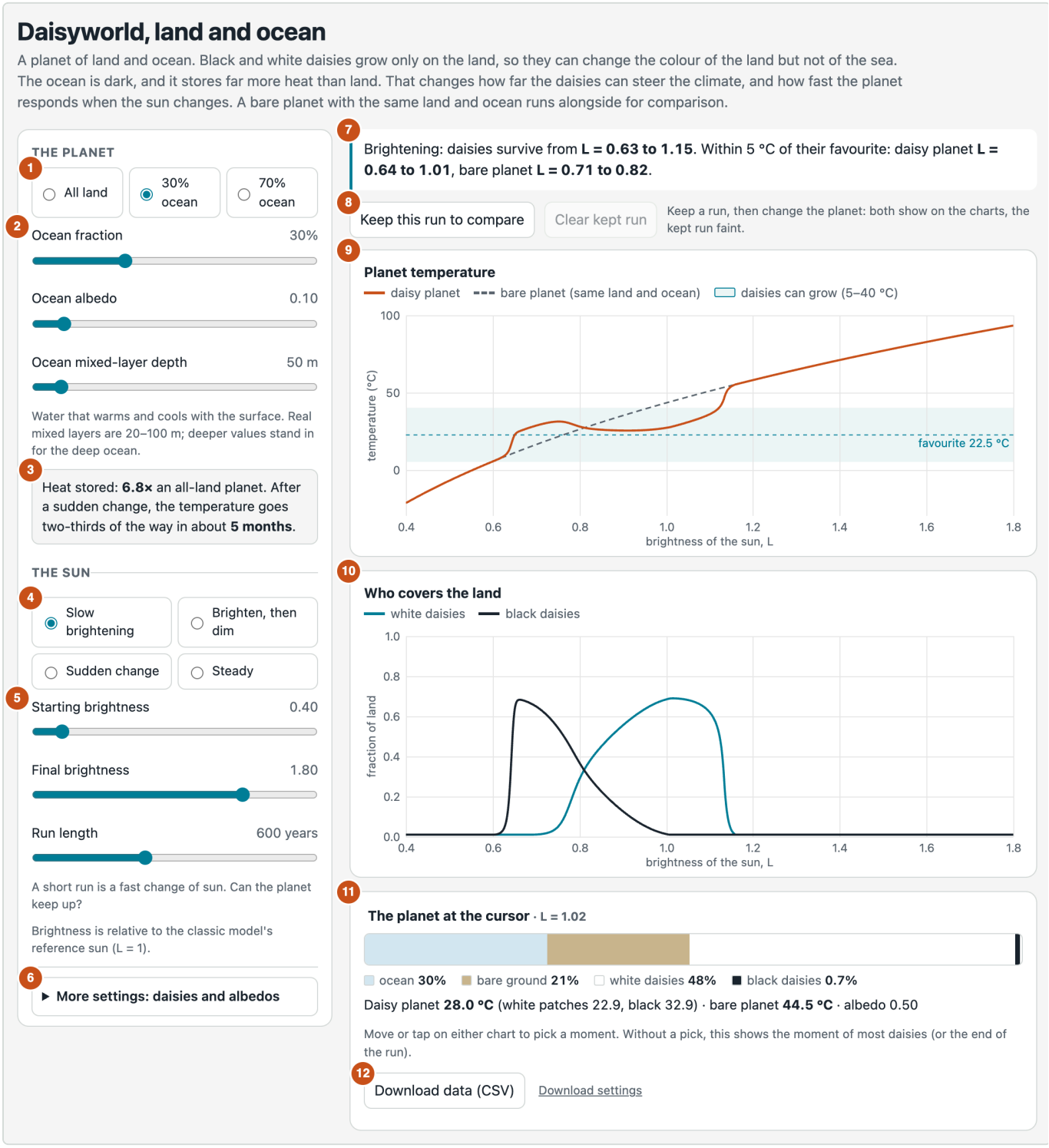
- **The daisies.** They live only on land and spread into bare ground. They grow best at 22.5 °C and cannot grow below 5 °C or above 40 °C. A fixed share of them dies every year. They need **years** to spread or to die back.
- **Patches.** This is the trick that makes Daisyworld work. A black patch is warmer than the planet's average and a white patch is cooler, by up to about 10 °C. Under a dim sun, the black patches are the ones warm enough to grow in. Black daisies spread, the planet darkens and it warms. Under a bright sun, the white patches are the comfortable ones. White daisies spread, the planet brightens and it cools. That is the whole regulating mechanism.
- **The heat store.** This is how much heat the planet must take in to warm by one degree. Land and air store little, so an all-land planet responds in weeks. The ocean's top layer, the **mixed layer** (50 m deep by default), stores about 20 times more, so the planet responds in months. Deeper settings stand in for slow mixing into the deep ocean, and the response takes years.
- **The bare planet.** A twin planet runs alongside, with the same land, ocean and sun but no daisies. It is the control experiment. Any difference between the two is the daisies' doing.

1.4 What the app does: four kinds of sun

Kind of sun (button)	What happens	Question it answers
Slow brightening (the default)	The sun brightens from L = 0.4 to 1.8 over 600 years, slowly enough that the planet stays close to balance.	Over what range of sun can the daisies keep the planet comfortable?
Brighten, then dim	The same, then back down again.	Once the daisies die, do they come back at the same sun?
Sudden change	The planet settles under one sun (L = 0.78). At year 0 the sun jumps (to 0.86), and the next 40 years are shown.	How fast does the planet respond, and what do the daisies do meanwhile?
Steady	The sun is fixed and the daisies start from a few seeds.	How the daisies take over. This handout doesn't use it, but try it.

In the first two, the horizontal axis of the charts is the **brightness of the sun**. In a sudden change it is **time**.

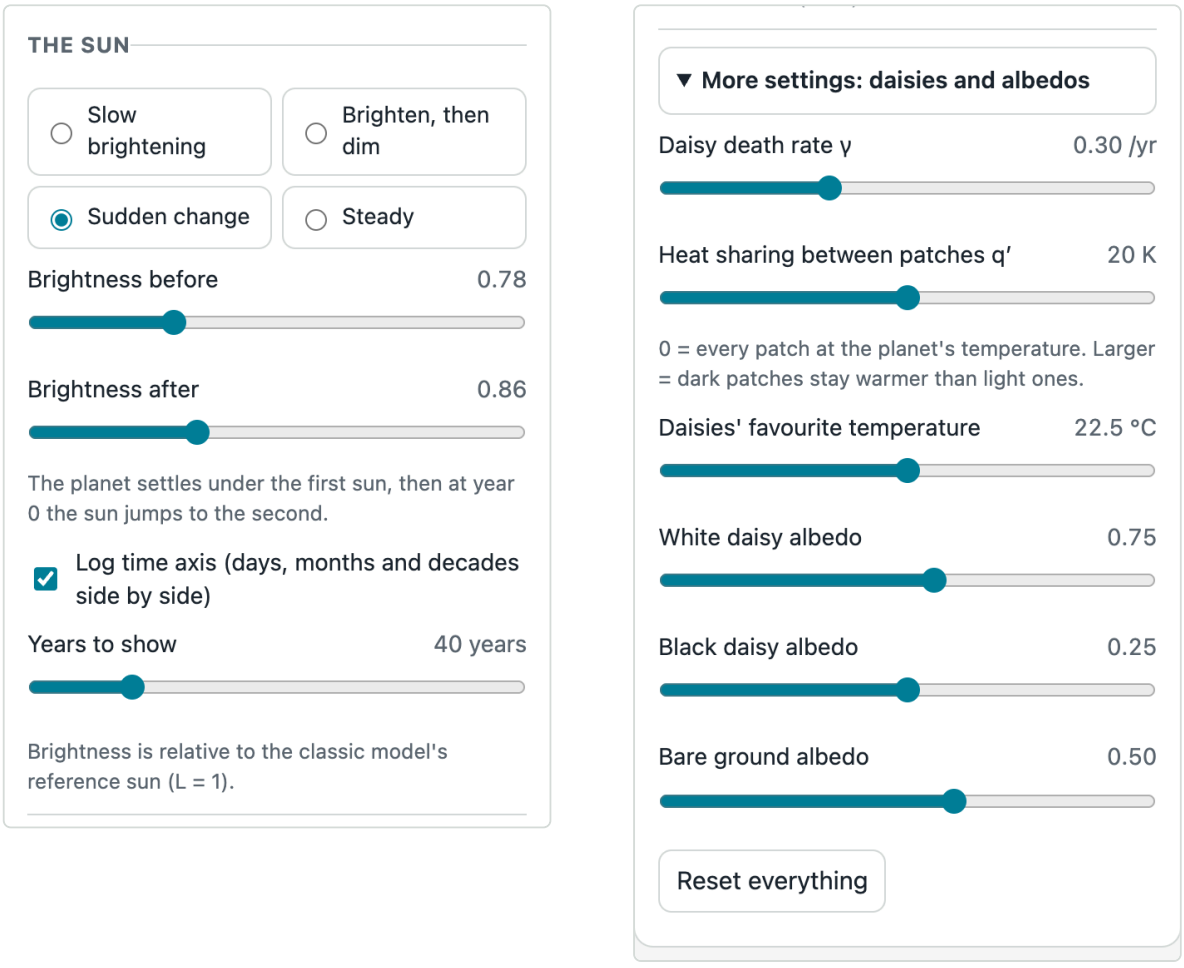
1.5 A tour of the screen



The app with 30% ocean under slow brightening. The numbers match the list below.

1. **Planet buttons**: all land, 30% ocean or 70% ocean.
2. **Planet sliders**: ocean fraction, ocean albedo and mixed-layer depth. The last two are dimmed when there is no ocean.
3. **Heat stored and response time.** These update as soon as you move a planet slider. They say how much heat the planet stores compared with an all-land planet, and how long it takes to go two-thirds of the way after a sudden change of sun.
4. **Kind of sun**: the four experiments in section 1.4.
5. **Sun controls.** These change with the kind of sun (pictures below). For the sweeps, they set the starting and final brightness and the run length. A short run is a fast change of sun.

6. **More settings: daisies and albedos.** This opens the daisies' death rate, how much heat the patches share, their favourite temperature and the albedos. It also holds **Reset everything**.
7. **Summary line.** It puts the main result in words. For a sweep, it gives where the daisies survive and where each planet stays within 5 °C of the daisies' favourite. For a sudden change, it gives how far and how fast each planet changes.
8. **Keep this run to compare.** It draws the current run faintly, so that after you change the planet both runs show. A kept run shows only while the kind of sun stays the same. **Clear kept run** removes it.
9. **Planet temperature.** The daisy planet is the solid orange line and the bare planet is the dashed grey line. The shaded band is where daisies can grow (5–40 °C), and the dotted line is their favourite, 22.5 °C.
10. **Who covers the land.** White daisies (teal) and black daisies (dark) as fractions of the **land**, not of the whole planet.
11. **The planet at the cursor.** Move over either chart, or tap it, to pick a moment. The strip shows how the whole planet is shared out: ocean, bare ground, white and black. Underneath are the planet's temperature, the temperatures of the white and black patches, the bare planet's temperature and the albedo. Without a pick, it shows the moment of most daisies in a sweep, or the end of a sudden change.
12. **Download data (CSV)** saves every recorded moment for a spreadsheet or Python. "Download settings" saves the parameters and a link that reopens the run.



*Left: the sun controls for **Sudden change**: brightness before and after, the **log time axis** box and the years to show. Right: **More settings** opened.*

Three more things worth knowing:

- **The address bar holds the settings.** Copy the link to save or share a run. Each run in Part 2 has a link that opens it.
- **Below the charts** are "Things to try" and "How the model works", the equations in one box.
- **On a phone** the controls come first and the charts sit below them.

Part 2. Ten runs to explore

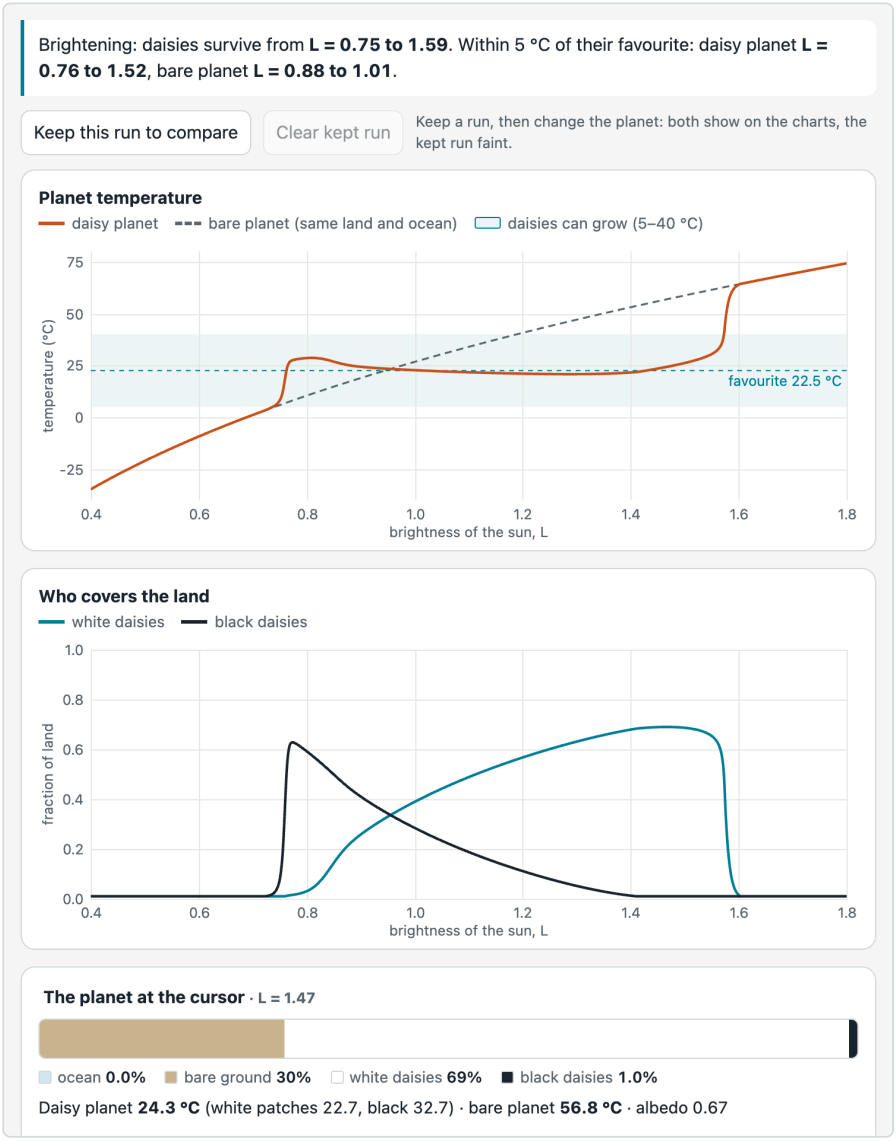
Each run says what to set, shows the screen, and asks a question for you to think about before you look. Then it lists what to look for and explains why it happens. Start from the defaults: **Reset everything** is under More settings, or just reload the page. The "open in the app" links set everything for you, except a kept run, which you have to make by hand.

Runs 1–5 ask **how much control** the daisies have. Runs 6–10 ask **how fast** the planet responds.

Run 1. All land, slow brightening: the classic Daisyworld

Set: All land · Slow brightening. (open in the app)

Before you look: the sun brightens steadily. Should the planet's temperature rise steadily too? What would the daisies do to it?



All land, slow brightening. The planet strip shows the moment of most daisies, at L = 1.47.

What to look for

- **The bare planet** (dashed) heats steadily as the sun brightens. It is within 5 °C of 22.5 °C only between *L* = 0.88 and 1.01.

- **The daisy planet** (solid) jumps up at $L \approx 0.75$, when daisies first grow. It then stays close to 22.5 °C from L = 0.76 to 1.52, a range of sun about six times wider than the bare planet manages. Over the middle of that range it even **cools slightly** as the sun brightens, from 23.4 to 20.9 °C between L = 0.95 and 1.33, while the bare planet heats from 22.9 to 48.8 °C.
- **The cover chart** tells the story. Black daisies come first, under the dim sun. White daisies take over as the sun brightens. At $L \approx 1.59$ the white daisies collapse and the daisy planet's temperature jumps up onto the bare planet's line.
- **At the moment of most daisies** (L = 1.47) white daisies cover 69% of the land. The planet is at 24.3 °C, while the bare planet is at 56.8 °C.

Why it happens

The balance. With the sun steady, the planet settles where the sunlight it absorbs equals the heat it radiates:

```
S · L · (1 − A) = σ T⁴
```

Here S = 917 W/m², A is the planet's albedo and σ is the Stefan–Boltzmann constant. A higher albedo means a cooler planet. The daisies' **only** lever on the climate is A.

The albedo of an all-land planet is the area-weighted average of its surfaces:

```
A = a_w · 0.75 + a_b · 0.25 + x · 0.5
```

Here *a_w* and *a_b* are the fractions of land covered by white and black daisies, and $x = 1 - a_w - a_b$ is bare ground.

The patches. Each colour of patch has its own temperature:

```
T_white = T + 20 · (A − 0.75)
T_black = T + 20 · (A − 0.25)
```

The 20 K (called *q′*) sets how little heat the patches share with each other. On a planet of albedo 0.5, a black patch sits 5 °C above the planet's temperature and a white patch 5 °C below it. Under a dim sun, the black patches reach the growing range first. Black daisies spread, A falls and T rises, which helps more black daisies grow. That is the jump at $L \approx 0.75$. As the sun brightens, the white patches become the comfortable ones. White daisies spread, A rises and the planet cools. The two colours pull in opposite directions and hold T where both can grow.

The growth. Each kind of daisy grows into bare ground at a rate set by its own patch's temperature, and dies at a fixed rate:

```
da_i/dt = a_i · [ x · β(T_i) − γ ]
β(T) = 1 − 0.003265 · (22.5 − T)²     (zero below 5 °C and above 40 °C)
```

Here γ = 0.3 per year. Daisies can never cover more than 70% of the land: when growth balances death, $x \cdot \beta = \gamma$, and β is at most 1, so bare ground x stays at 0.3 or more. At $L \approx 1.5$ the white daisies are near that limit. The dial has reached its end stop. A little more sun warms the white patches beyond what the daisies can stand. Some die, the albedo falls, the planet warms, and more die. This runaway ends on the bare planet's line.

Run 2. Add ocean: 30% and 70%

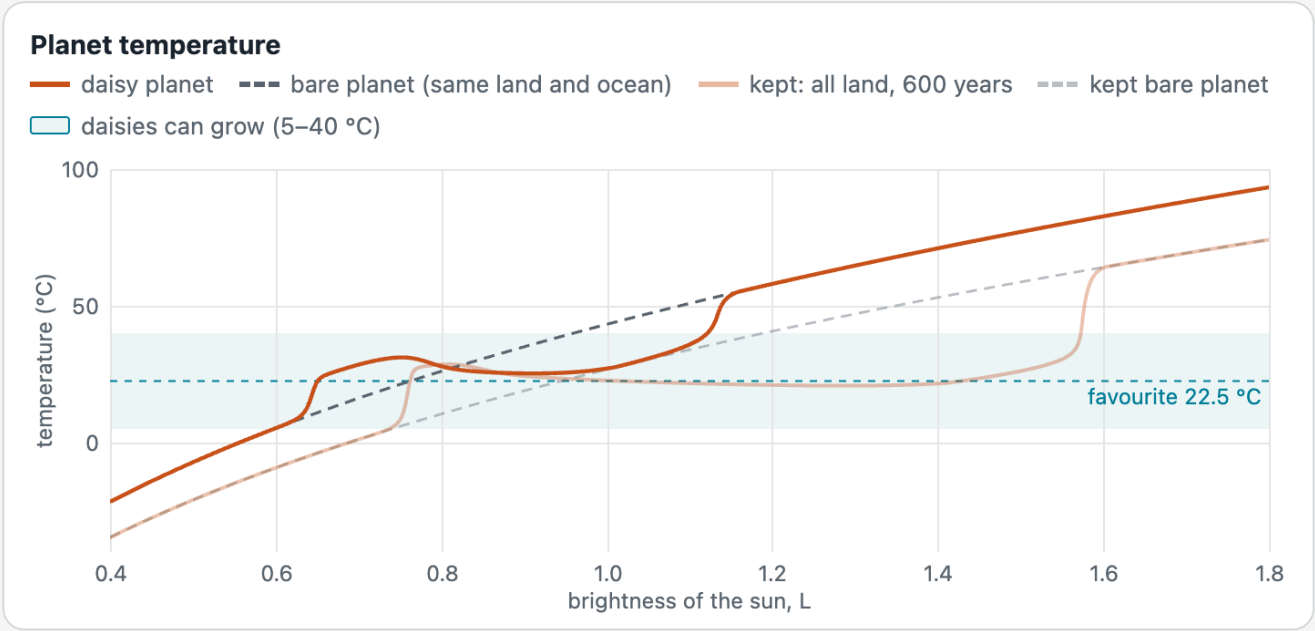
Set: Run 1, then press **Keep this run to compare**. Then choose 30% ocean (open in the app), and then 70% ocean (open in the app).

Before you look: daisies can't grow on the ocean, and the ocean is dark. Will the range of sun over which the daisies keep the planet comfortable get wider, get narrower, or move?

Brightening: daisies survive from **L = 0.63 to 1.15**. Within 5 °C of their favourite: daisy planet **L = 0.64 to 1.01**, bare planet **L = 0.71 to 0.82**.

Keep this run to compare | Clear kept run

Kept: all land, 600 years. It shows while the kind of sun is the same.



30% ocean (solid) against the kept all-land run (faint).



70% ocean against the kept all-land run. The cursor is at L = 0.85: see Run 4.

What to look for

Slow brightening	All land	30% ocean	70% ocean
Daisies survive	*L* = 0.75–1.59	0.63–1.15	0.52–0.89
Daisy planet within 5 °C of 22.5 °C	0.76–1.52	0.64–1.01	0.55–0.61
Bare planet within 5 °C of 22.5 °C	0.88–1.01	0.71–0.82	0.57–0.65

- **The window moves to a dimmer sun.** The ocean planets are warmer under the same sun.
- **The window narrows.** With 30% ocean, the regulated range is less than half as wide (0.36 against 0.76 in *L*). The regulated temperature also runs warmer, around 25–30 °C instead of 21–24 °C.
- **With 70% ocean the two curves nearly overlap.** The daisies grow over *L* = 0.52–0.89, but the daisy planet stays comfortable only over 0.55–0.61, which is no wider than the bare planet's range. **The daisies live on this planet; they don't run it.**

Why it happens

With a fraction *f* of the planet as land and 1 – *f* as ocean, the albedo becomes:

```
A = f · (a_w · 0.75 + a_b · 0.25 + x · 0.5) + (1 − f) · 0.1
```

This one line is the whole story of **leverage**. On land, daisies can move the albedo anywhere between 0.25 (all black) and 0.75 (all white), a swing of 0.5. On the whole planet that swing is 0.5 × f: 0.5 with all land, 0.35 with 30% ocean and only 0.15 with 70% ocean. The ocean's share is fixed, and it is dark. More ocean gives the daisies a smaller dial, and it sets the planet warmer to begin with.

Run 3. Two reasons, pulled apart: a bright ocean

More ocean changes two things at once: there is less land to recolour, **and** the planet gets darker. The ocean-albedo slider separates them.

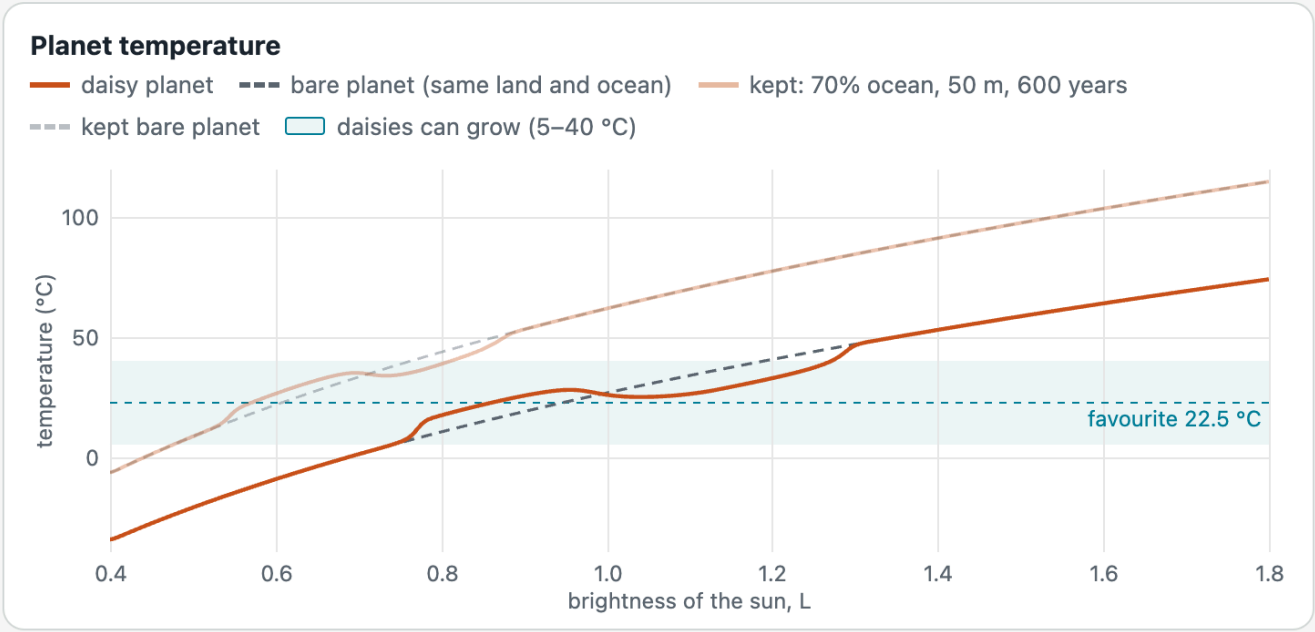
Set: 70% ocean, then **Keep**. Then set the ocean albedo to 0.5, the same as bare ground. (open in the app)

Before you look: of the two changes in Run 2 (the window moving and the window narrowing), which one should disappear when the ocean is as bright as bare ground?

Brightening: daisies survive from **L = 0.75 to 1.30**. Within 5 °C of their favourite: daisy planet **L = 0.80 to 1.13**, bare planet **L = 0.88 to 1.01**.

Keep this run to compare | Clear kept run

Kept: 70% ocean, 50 m, 600 years. It shows while the kind of sun is the same.



70% ocean with a bright ocean (albedo 0.5, solid) against the real, dark ocean (0.1, faint).

Ocean albedo 0.5	All land	30% ocean	70% ocean
Daisies survive	0.75–1.59	0.75–1.44	0.75–1.30
Daisy planet within 5 °C	0.76–1.52	0.77–1.32	0.80–1.13

- The window **no longer moves**: it starts at $L \approx 0.75$ every time. So the shift to a dimmer sun came from the ocean being **dark**.
- The window **still narrows**: its top end falls from 1.59 to 1.30. That narrowing is **leverage** alone.

The real ocean is the darkest large surface on Earth, which is why it takes in about half of all the sunlight Earth absorbs. In the model, that darkness warms the planet and moves life's window.

Run 4. Surviving is not regulating

Set: the 70% ocean run from Run 2. Put the cursor at L = 0.85, near the hot end of the daisies' range (see the second picture in Run 2).

Before you look: the readout says the planet is at 44.8 °C, above the 40 °C limit for daisies. Yet white daisies cover 47% of the land. How can they?

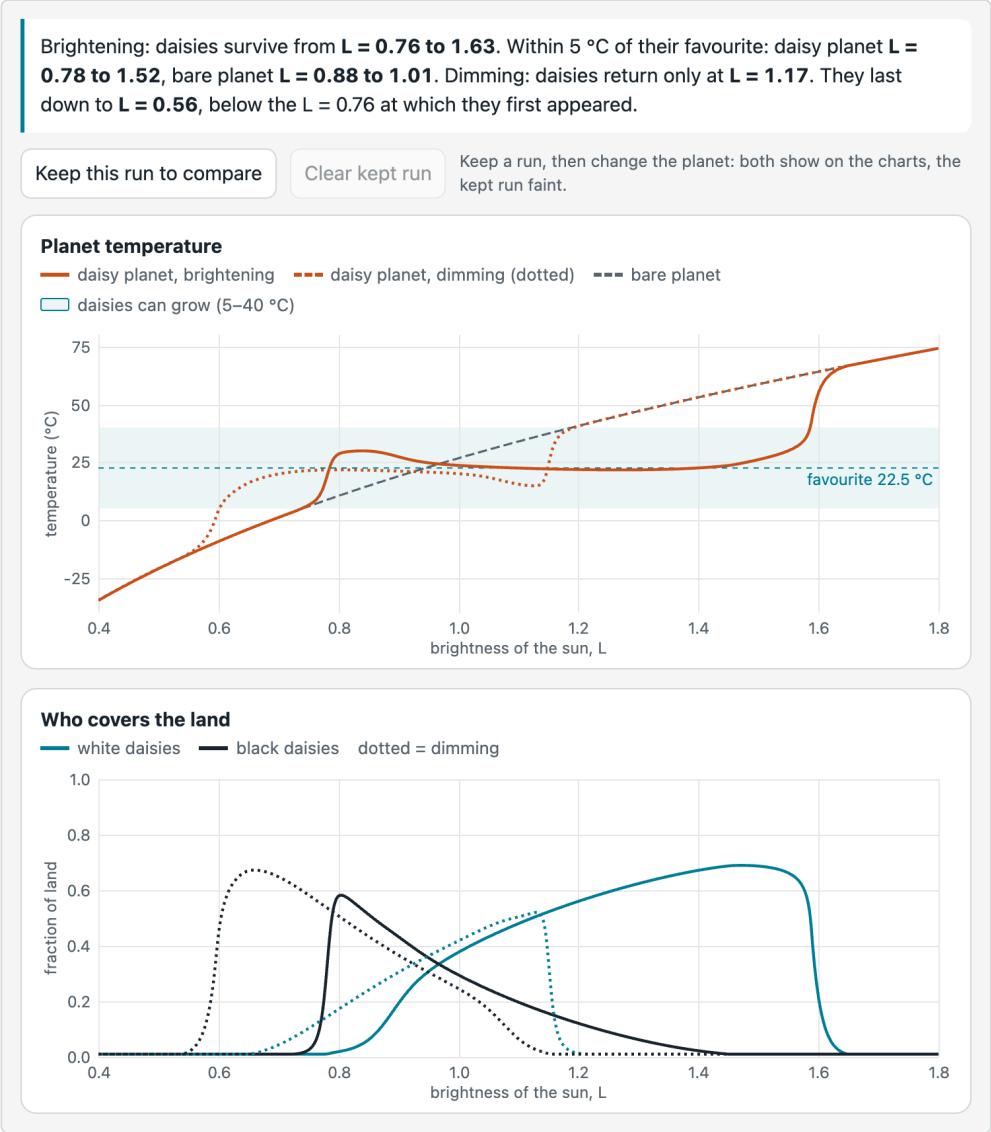
Near the hot edge	*L*	Planet	White patches	Planet albedo
All land	1.55	30.4 °C	28.6 °C	0.66
30% ocean	1.10	35.8 °C	30.6 °C	0.49
70% ocean	0.85	44.8 °C	34.9 °C	0.25

Why. Look again at $T_white = T + 20 \cdot (A - 0.75)$. On a dark ocean planet ($A \approx 0.25$), white patches sit 10 °C below the planet's temperature. The white daisies have found a cool niche and keep growing in it. But land is only 30% of this planet, and white daisies cover less than half of it (14% of the planet), so they can't cool the rest. Life can **find** a habitable spot on a planet it can't **make** habitable.

Run 5. Brighten, then dim: the daisies don't come straight back

Set: All land · Brighten, then dim. (open in the app)

Before you look: once the daisies have died under a bright sun, will they come back as soon as the sun dims to where they died?



All land, brighten then dim. The dotted lines are the way down.

What to look for. On the way up the daisies die at $L \approx 1.63$. On the way down (dotted) they come back only at $L = 1.17$. Between those two values the same sun gives two different planets, one with daisies and one without. That is **hysteresis**. The numbers on the way up differ slightly from Run 1 (0.76–1.63 rather than 0.75–1.59) because this sweep is twice as fast: 300 years up and 300 down. That is a first hint that how fast the sun changes matters (Runs 6–10).

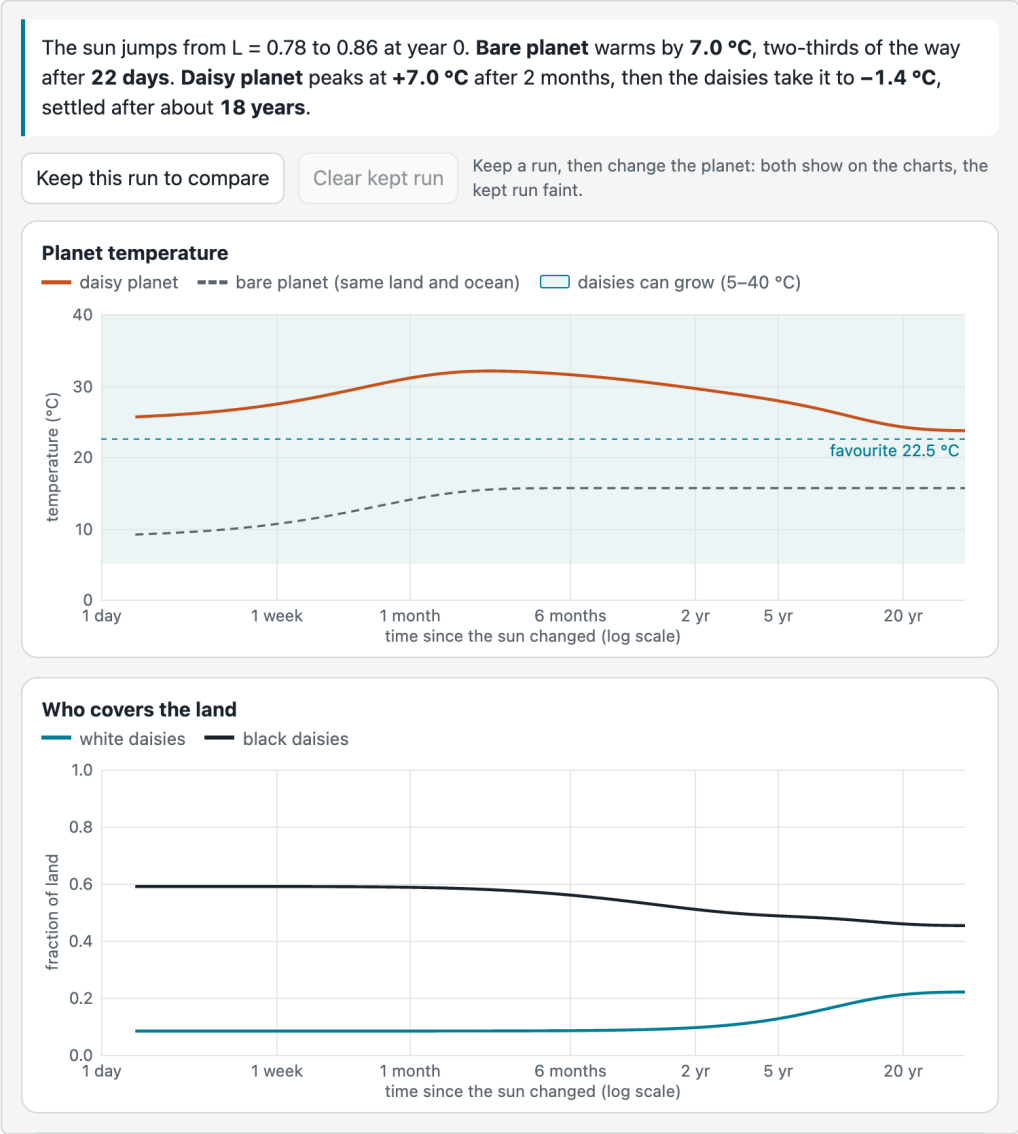
Brighten, then dim	All land	30% ocean	70% ocean
Daisies die on the way up at	1.63	1.18	0.91
Daisies come back on the way down at	1.17	0.97	0.80
Width of the loop	0.45	0.21	0.11

Why. A planet without daisies is darker than one covered in white daisies, so it is hotter. The sun has to dim a long way before even the white patches are cool enough for seeds to take hold. With less land, the daisies change the planet less, so the planet with daisies and the planet without them differ less, and the loop shrinks. That is one more sign of less control.

Run 6. Sudden change, all land: a spike, then a correction

Set: All land · Sudden change · tick the **log time axis**. (open in the app)

Before you look: the sun jumps. How long will the bare planet take to warm: days, months or years? Will the daisy planet warm as much?



All land: the sun jumps from L = 0.78 to 0.86 at year 0. On the log axis, days, months and decades get equal room.

What to look for

- **Before the change**, black daisies cover 59% of the land and hold the daisy planet at 25.0 °C, while the bare planet sits at 8.7 °C.
- **The bare planet** warms by 7.0 °C. It gets two-thirds of the way in **22 days** and is flat after a few months.
- **The daisy planet** does just what the bare planet does for the first two months: it jumps by 7 °C. Then, over years, the white daisies spread (from 8% to 22% of the land) and the black daisies thin out (from 59% to 45%). The planet comes back, and settles after about 18 years at 1.4 °C **cooler** than before.

Why it happens

The energy budget, with storage. When the planet is out of balance, the difference goes into (or comes out of) its heat store, *C*:

$$C \cdot \frac{dT}{dt} = S \cdot L \cdot (1 - A) - \sigma T^4$$

Absorbed minus emitted equals the rate at which heat is stored. At the jump, the absorbed sunlight rises by 46 W/m² at once, but the emitted heat catches up only as the planet warms. The planet in this model has **no greenhouse**: it radiates straight from its surface.

The heat store is shared between land and ocean:

```
C = f · C_land + (1 − f) · ρc · h
```

Here *C_land* = 10^7 J m^{-2} K^{-1} stands for land plus the air above it, about 2.5 m of water's worth. ρc = 4.1 × 10^6 J m^{-3} K^{-1} is for sea water, and *h* is the mixed-layer depth. At 50 m, the ocean stores about 20 times as much heat per square metre as land.

The response time. After a small sudden change, a planet goes about two-thirds of the way to its new temperature in

```
τ = C / λ,      λ = 4σT³ ≈ 5.4 W m⁻² K⁻¹
```

Here λ is how much more heat the planet emits for each degree it warms. For all land, τ ≈ 21 days, and the run measures 22 days. In lecture, λ was the Planck feedback for Earth, 3.8 W m^{-2} K^{-1}, which includes the greenhouse effect. This model has no greenhouse, so λ is larger and τ is shorter: 50 m of water gives about **1.2 years here** against **1.7 years** in lecture.

Heat capacity doesn't change where the planet ends up. Set d*T*/d*t* = 0 and *C* drops out. It changes only **how fast** the planet gets there.

The spike. The temperature responds in weeks and the daisies in years. So for the first months the daisies are effectively frozen, and the daisy planet warms just like the bare one. **Fast physics plus slow biology gives an overshoot.** Ending cooler under a brighter sun is not a bug. Run 1 showed it already: in the middle of its regulated range, the daisy planet cools slightly as the sun brightens.

Run 7. Sudden change, 30% ocean: the same push, a bigger store

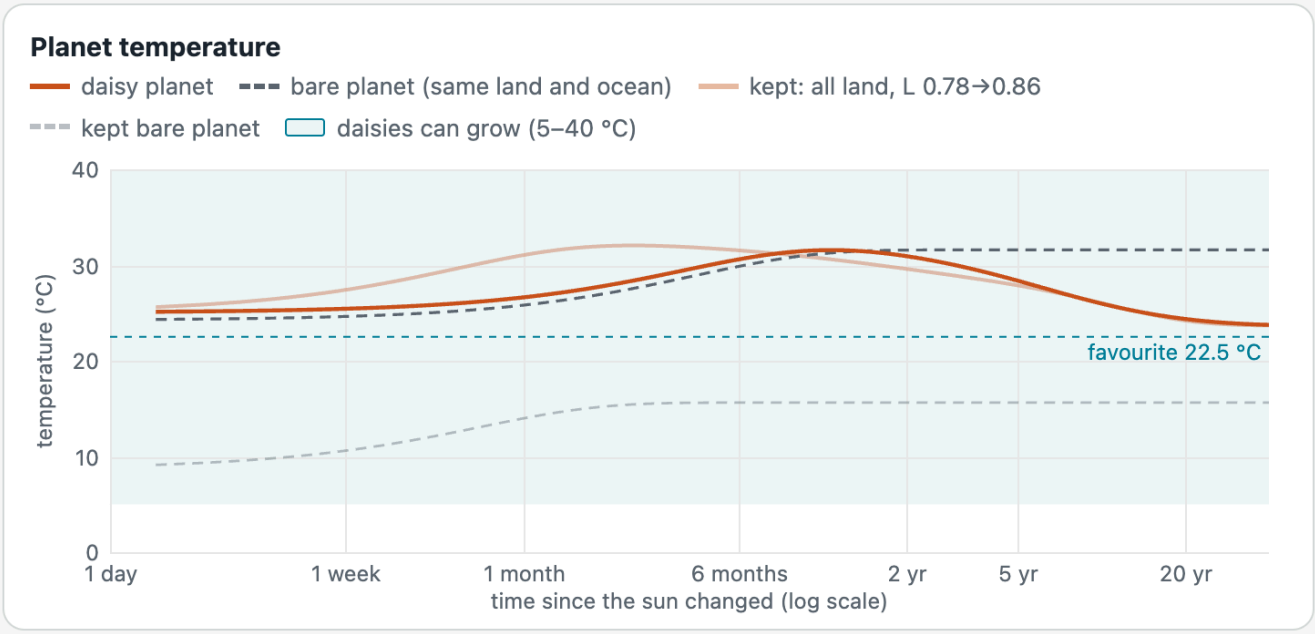
Set: Run 6 and press **Keep**. Then choose 30% ocean. (open in the app)

Before you look: the response-time readout (item 3 of the tour) now says about 5 months. Will the daisy planet's spike be bigger or smaller than on the all-land planet?

The sun jumps from L = 0.78 to 0.86 at year 0. **Bare planet** warms by **7.3 °C**, two-thirds of the way after **4 months**. **Daisy planet** peaks at **+6.5 °C** after 13 months, then the daisies take it to **−1.3 °C**, settled after about **20 years**.

Keep this run to compare | Clear kept run

Kept: all land, L 0.78→0.86. It shows while the kind of sun is the same.



30% ocean with a 50 m mixed layer (solid) against all land (faint).

What to look for. The bare planet now warms by 7.3 °C, two-thirds of the way after about **4 months** (127 days) instead of 22 days. The daisy planet peaks **later** (after about a year) and **lower** (+6.5 °C). It then settles at −1.3 °C after about 20 years.

Why: an accountant's view. The sun jumps, and the planet suddenly absorbs 46 W/m² more than it emits, the same surplus on both planets. That surplus has to fill the store until the planet is warm enough to emit it away. Both bare planets end up about 7 °C warmer, but the energy needed to get there is $C \times \Delta T$: about 7×10^7 J/m² for all land against 5×10^8 J/m² with 30% ocean. The bigger store takes longer to fill.

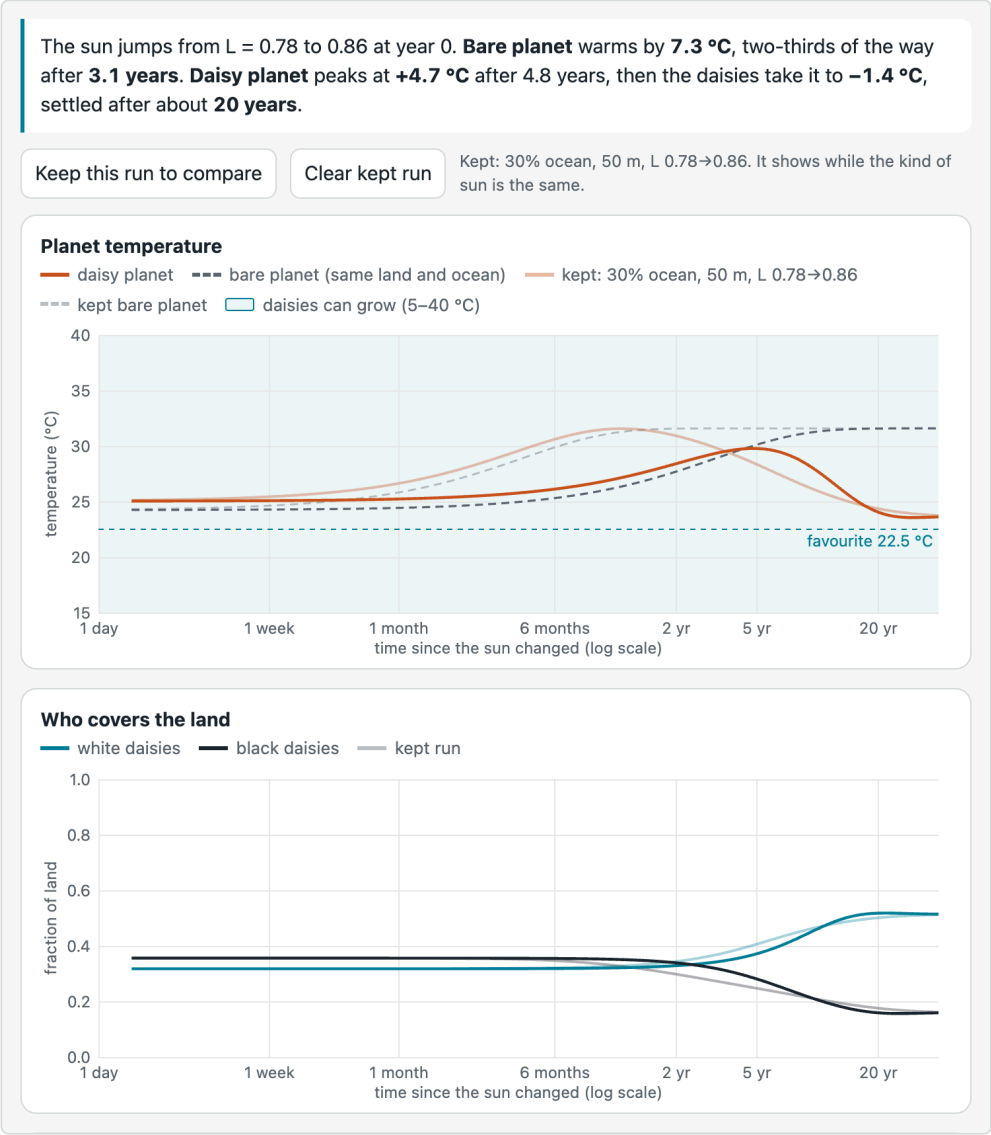
Planet	Heat stored (× all land)	τ
All land	1.0	21 days
30% ocean, 10 m	1.9	41 days
30% ocean, 50 m	6.8	146 days (about 5 months)
30% ocean, 200 m	25	1.5 years
30% ocean, 500 m	62	3.6 years
70% ocean, 50 m	15	0.85 years (about 10 months)
70% ocean, 200 m	58	3.4 years

The measured times come out a little shorter than τ (127 days against 146 days here), because a planet emits more steeply as it warms than the straight-line λ assumes.

Run 8. A deeper ocean buys the daisies time

Set: Run 7 (30% ocean, 50 m) and press **Keep**. Then set the mixed-layer depth to 500 m. (open in the app)

Before you look: with ten times the water, will the daisy planet's peak warming be bigger or smaller? Will it end up somewhere different?



30% ocean with a 500 m mixed layer (solid) against 50 m (faint).

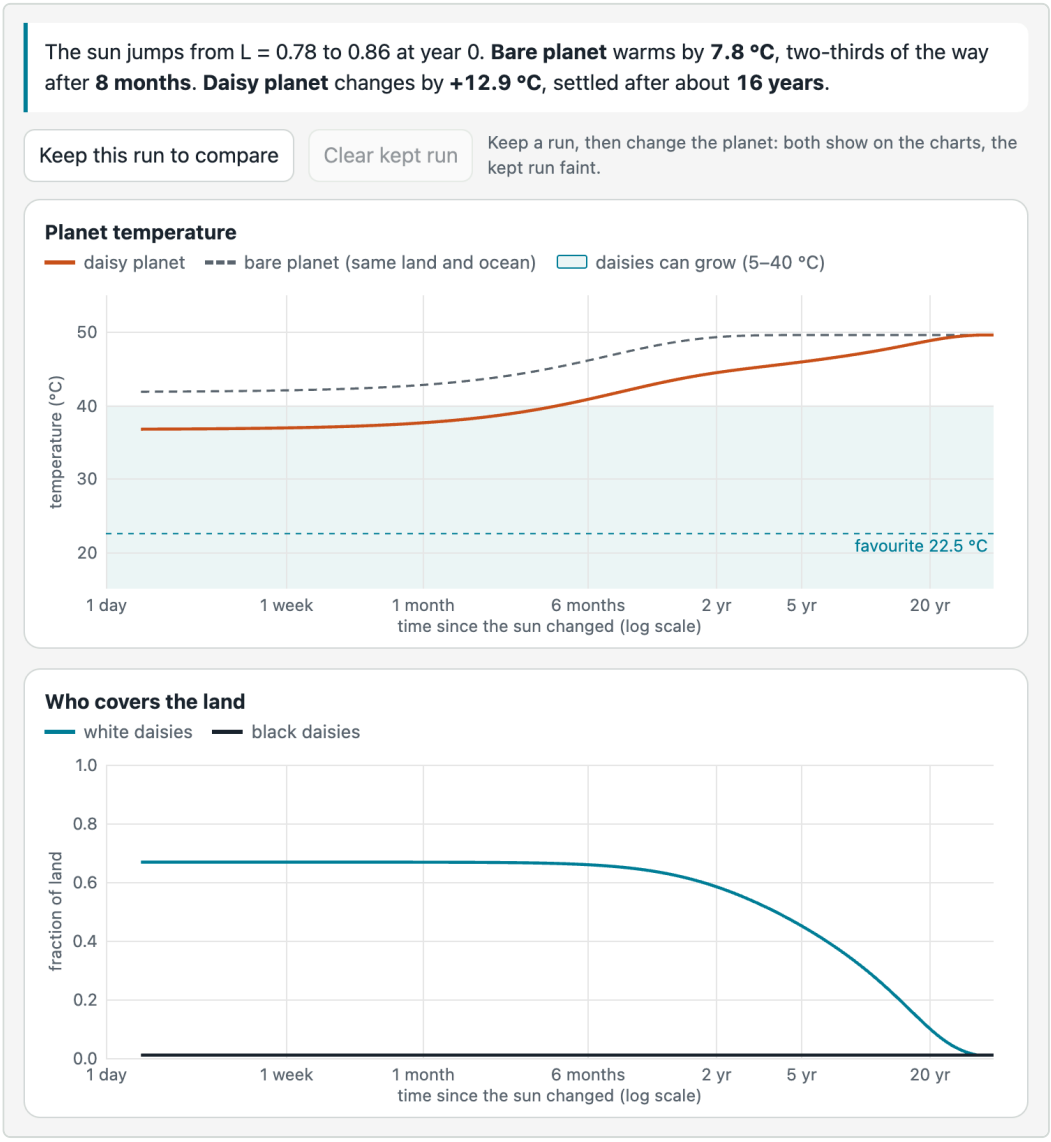
30% ocean	Bare planet: two-thirds of the way after	Daisy planet peak	Daisy planet at the end
50 m	about 4 months	+6.5 °C after about 1 year	−1.3 °C
200 m	about 15 months	+5.6 °C after about 2.7 years	−1.3 °C
500 m	about 3 years	+4.7 °C after about 5 years	−1.4 °C

Why. The deeper ocean slows the warming, so the heat arrives over years rather than months. That gives the daisies, which also work over years, time to respond before the heat has arrived, and the peak is lower. The end point doesn't move: about −1.3 °C after about 20 years, whatever the depth. **Heat capacity changes the path, not the destination.**

Run 9. Sudden change, 70% ocean: a slow feedback takes over

Set: 70% ocean · Sudden change · log time axis. (open in the app)

Before you look: the response-time readout says about 10 months. When will this planet's warming be over?



70% ocean, sudden change. The bare planet is done within a couple of years; the daisy planet isn't.

What to look for

- **Before the change** this planet is already hot. The daisy planet is at 36.7 °C and survives only through white daisies (67% of the land) in their cool patches. The bare planet is at 41.8 °C.
- **The bare planet** does what you'd now expect: +7.8 °C, two-thirds of the way in about 8 months.
- **The daisy planet keeps warming for decades.** The brighter sun pushes the white patches past their limit, and the white daisies die back over the next 10–20 years. As they go, the planet darkens and warms further. It ends 12.9 °C warmer, at 49.6 °C, which is exactly the bare planet's temperature, because no daisies are left: the daisy planet has **become** a bare planet.

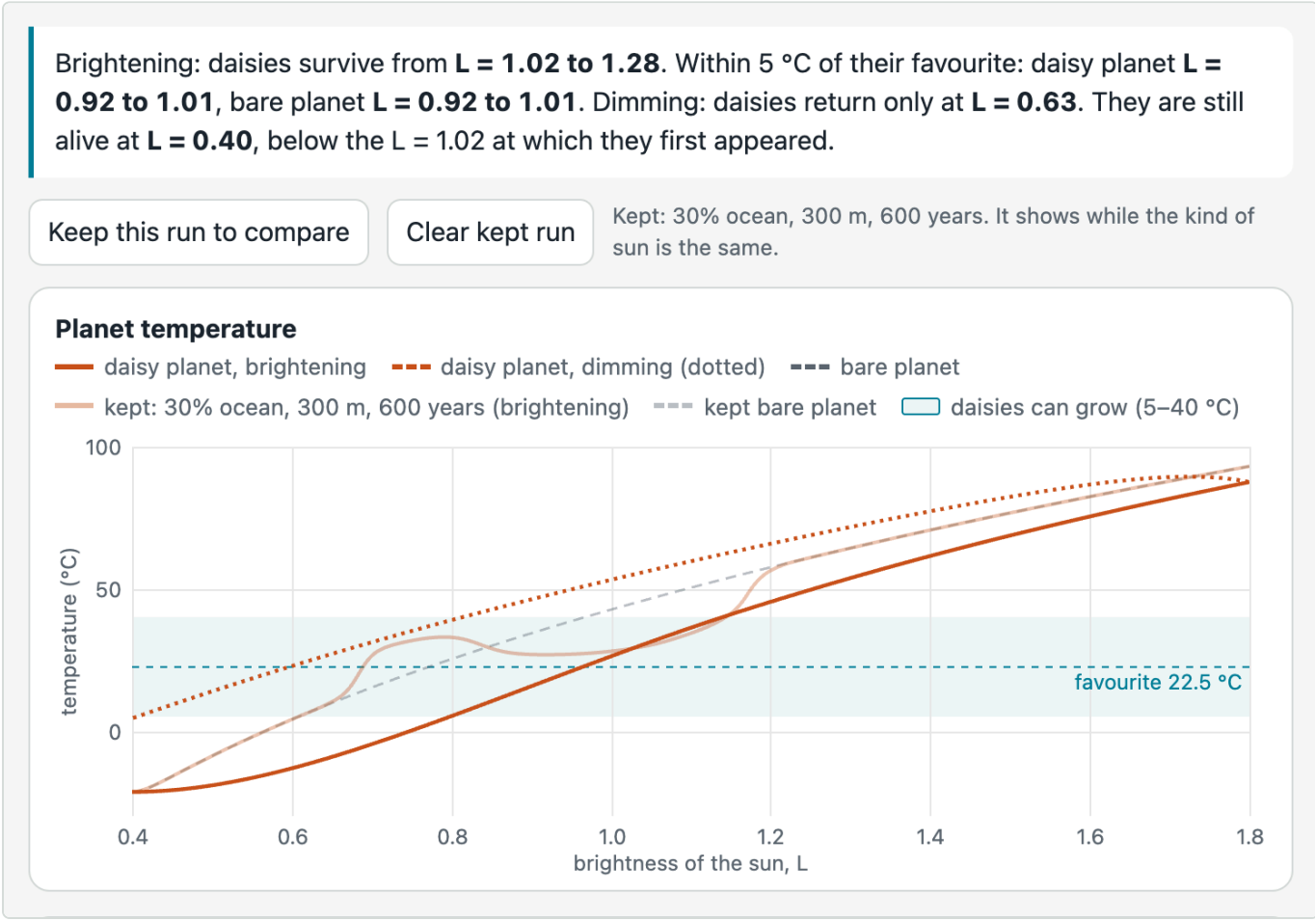
Why. The heat from the sun arrives within a year, but the warming lasts decades, because a **slow feedback** (the daisies dying back) sets the pace, not the ocean. With a 200 m ocean the end point is the same (+12.8 °C); only the path differs.

Try next: a gentle change inside this planet's narrow comfortable window, brightness before 0.58 and after 0.62 (open in the app). The daisy planet then warms by 4.8 °C, the same as the bare planet (4.9 °C). The daisies make no difference, and all that's left is the ocean's delay: about 10 months with 50 m, about 3 years with 200 m.

Run 10. Too fast to follow

Set: 30% ocean · mixed layer 300 m · Brighten, then dim · run length 600 years, and press **Keep**. Then set the run length to 30 years. (open in the app)

Before you look: the response time for this planet is a few years. If the sun brightens and dims over 30 years, can the planet keep up?



30% ocean, 300 m: brighten and dim over 30 years (solid and dotted) against 600 years (faint).

What to look for. On the fast run, the planet on the way up (solid) runs far **colder** than on the slow run, and on the way down (dotted) far **hotter**: the temperature trails the sun. The daisies appear only at L = 1.02 on the way up, and on the way down they hang on all the way to L = 0.4.

Brighten, then dim, over 30 years	Bare planet lags its balance by up to	Daisies alive on the way up	Return on the way down
All land	1.3 °C	0.95–1.44	0.96
30% ocean, 50 m	6.2 °C	0.86–1.16	0.75
30% ocean, 300 m	21 °C	1.02–1.28	0.63
30% ocean, 300 m, over 600 years	2.1 °C	0.65–1.20	0.96

Why. With all land, the temperature keeps up (it lags by 1.3 °C at most), but the **daisies** don't: they start at L = 0.95 instead of 0.76, because they need years to spread. With a 300 m ocean the **temperature** itself falls 21 °C behind its balance. This is the transient version of hysteresis: the faster the forcing, the more the planet's state depends on its history.

Putting the runs together: three clocks

The log time axis shows the model's three clocks:

Part	Responds in	Set by
Land and air	weeks	a small heat capacity
Ocean mixed layer	months to years	heat capacity: each metre of water stores 0.4 times what land and air do, so 50 m stores 20 times as much
Daisies	years to decades	growth and death rates

Most of what happens after a sudden change comes from how these clocks compare. When the temperature clock is much faster than the daisy clock, you get a spike (Run 6). When the two get closer, the spike shrinks (Run 8). When the daisy clock runs the show, the planet responds as slowly as the biology (Run 9).

The model in full

For reference, here is everything the app computes:

```
albedo     A = f · (a_w · A_w + a_b · A_b + x · A_g) + (1 − f) · A_o
energy     C · dT/dt = S · L · (1 − A) − σ T⁴
storage    C = f · C_land + (1 − f) · ρc · h
patches    T_w = T + q′ · (A − A_w),   T_b = T + q′ · (A − A_b)
growth     β(T) = 1 − 0.003265 · (T_opt − T)²   (zero outside T_opt ± 17.5 °C)
daisies    da_i/dt = a_i · [ x · β(T_i) − γ ],   x = 1 − a_w − a_b
```

Symbol	Meaning	Default
f	land fraction (1 – ocean fraction)	1, 0.7 or 0.3
A_w, *A_b*, *A_g*, *A_o*	albedo of white daisies, black daisies, bare ground, ocean	0.75, 0.25, 0.5, 0.1
S	sunlight at *L* = 1	917 W/m²
C_land	heat capacity of land plus air	10^7 J m^{-2} K^{-1}
ρc, *h*	heat capacity of sea water per m³, mixed-layer depth	4.1 × 10^6 J m^{-3} K^{-1}, 50 m
q′	how much warmer dark patches stay than light ones	20 K
T_opt	daisies' favourite temperature	22.5 °C
γ	daisy death rate	0.3 per year

Daisy cover never falls below 1% of the land (seeds). The bare planet uses the same equations with no daisies. Growth, albedos, *S*, γ and *q′* follow Watson and Lovelock (1983). The ocean and the heat store are simple teaching choices.

Part 3. What to take away

3.1 Six points

1. **Leverage.** Daisies can only change the colour of land. With 70% ocean they still grow, but they barely move the planet's temperature. *Surviving is not the same as regulating* (Runs 2 and 4).
2. **A dark ocean shifts everything.** The ocean reflects little sunlight, so an ocean planet is warmer under the same sun, and life's window moves to a dimmer sun (Run 3).
3. **Heat capacity sets the response time.** After a sudden change the bare planet always ends up about 7–8 °C warmer, but it takes about 3 weeks with all land, 4 months with 30% ocean and 8 months with 70% ocean. A deeper mixed layer makes it years. $\tau = C/\lambda$: the bigger the store, the longer it takes to fill (Runs 6–8).
4. **A transient is a time out of balance.** While the planet catches up, it absorbs more than it emits, and the difference goes into storage. Earth today is in such a transient: it absorbs about 1 W/m² more than it emits, and more than 90% of the excess goes into the ocean.
5. **Fast physics plus slow biology gives an overshoot.** The temperature jumps before the daisies can react, and the daisies pull it back over years. The more slowly the temperature rises (a deeper ocean), the more time the daisies have and the smaller the peak (Runs 6 and 8).
6. **The planet's response time isn't only C/λ.** A slow feedback can stretch it: with 70% ocean the heat arrives within a year, but the planet keeps warming for decades as the daisies die back (Run 9). And if the sun changes faster than the planet can follow, the planet lags behind (Run 10).

3.2 What the model leaves out

A model is useful partly because you know where it breaks.

- **One temperature.** On Earth, land and ocean have their own temperatures and their own response times: the yearly range of temperature is tens of degrees inside continents and only a few degrees over the open ocean. Here, adding ocean slows the land down too, because they share one T. This is the biggest simplification.
- **No transport.** On Earth, no place is in energy balance on its own, and the atmosphere and ocean currents carry heat from where there is a surplus to where there is a deficit. A one-box model has no places, so it can't move anything.
- **No sea ice.** The real ocean's albedo jumps from about 0.06 to about 0.6 when it freezes, a strong feedback. Here the ocean stays dark at every temperature.
- **One slab.** The real ocean exchanges heat with its deep layers over decades to centuries, so it responds in two stages, fast and then very slow.
- **No greenhouse, no clouds, no weather.** These keep the model simple.
- **Daisies are a parable.** Real vegetation does change albedo (dark forest against snow), but on Earth that effect is small next to clouds, ice and the ocean.